

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

## CO5 - ASSESSMENT TOOL 2 – Coding Challenge

|                         |                                                                                      |                      |                                      |
|-------------------------|--------------------------------------------------------------------------------------|----------------------|--------------------------------------|
| <b>Course Code:</b>     | DSA0405                                                                              | <b>Course Title:</b> | FUNDAMENTALS OF DATA SCIENCE         |
| <b>CO Assessed:</b>     | CO5 – Coding Challenge                                                               | <b>Assessment:</b>   | Assessment Tool 2 – Coding Challenge |
| <b>Weightage:</b>       | 40% of CIA                                                                           | <b>Total Marks:</b>  | 100                                  |
| <b>CO5 Statement:</b>   | To understand the concept of supervised and unsupervised methods in machine learning |                      |                                      |
| <b>Student Name:</b>    | Mohamed Bazith.M                                                                     | <b>Reg. No.:</b>     | 192472052                            |
| <b>Section / Batch:</b> | 2024                                                                                 | <b>Date:</b>         | 27/8/2026                            |

### Code of Conduct

I Mohamed Bazith.M (192472052) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

**Signature of the Student: Mohamed Bazith.M**

| Criteria                       | Weightage | Level 4 – Exemplary                                              | Level 3 – Proficient                           | Level 2 – Developing                    | Level 1 – Beginning                                  | Marks |
|--------------------------------|-----------|------------------------------------------------------------------|------------------------------------------------|-----------------------------------------|------------------------------------------------------|-------|
| <b>Understanding of Case</b>   | 25        | Thorough understanding; clearly identifies all key issues        | Good understanding with most issues identified | Basic understanding; some issues missed | Poor understanding; problem not identified correctly |       |
| <b>Application of Concepts</b> | 25        | Effectively applies relevant concepts to analyze the case deeply | Applies appropriate concepts with minor gaps   | Limited application of concepts         | Fails to apply relevant concepts                     |       |
| <b>Analysis</b>                | 20        | In-depth analysis with strong reasoning and insights             | Good analysis with logical reasoning           | Basic analysis with limited depth       | Weak or no analysis; lacks reasoning                 |       |
| <b>Solution Design</b>         | 20        | Innovative, feasible solutions with strong justification         | Logical solutions with adequate justification  | Basic solutions with weak justification | Incorrect or no solution provided                    |       |
| <b>Clarity</b>                 | 10        | Clear, well-structured, and easy to understand                   | Organized and understandable presentation      | Limited clarity and structure           | Poor clarity; difficult to understand                |       |

**1. Extend your CART implementation to support entropy/information gain as an alternative splitting criterion, controllable via a parameter. Run both on the same dataset and report where the resulting trees differ in structure.**

**Code:**

```
from sklearn.datasets import load_iris
from sklearn.tree import DecisionTreeClassifier, export_text

X, y = load_iris(return_X_y=True)

gini_tree = DecisionTreeClassifier(criterion="gini", random_state=42)
gini_tree.fit(X, y)

entropy_tree = DecisionTreeClassifier(criterion="entropy", random_state=42)
entropy_tree.fit(X, y)

print("TREE USING GINI INDEX (CART):")
print(export_text(gini_tree, feature_names=load_iris().feature_names))

print("\nTREE USING ENTROPY / INFORMATION GAIN:")
print(export_text(entropy_tree, feature_names=load_iris().feature_names))

gini_structure = export_text(gini_tree, feature_names=load_iris().feature_names)
entropy_structure = export_text(entropy_tree, feature_names=load_iris().feature_names)

print("\nCOMPARISON:")
if gini_structure == entropy_structure:
    print("Both trees have the same structure.")
else:
    print("The trees differ in structure.")
    print("Gini selects splits using impurity reduction.")
    print("Entropy selects splits using Information Gain.")
```

**Output:**

```

TREE USING GINI INDEX (CART):
|--- petal length (cm) <= 2.45
|   |--- class: 0
|--- petal length (cm) > 2.45
|   |--- petal width (cm) <= 1.75
|   |   |--- petal length (cm) <= 4.95
|   |   |   |--- petal width (cm) <= 1.65
|   |   |   |   |--- class: 1
|   |   |   |   |--- petal width (cm) > 1.65
|   |   |   |   |   |--- class: 2
|   |   |   |--- petal length (cm) > 4.95
|   |   |   |   |--- petal width (cm) <= 1.55
|   |   |   |   |   |--- class: 2
|   |   |   |   |   |--- petal width (cm) > 1.55
|   |   |   |   |   |   |--- petal length (cm) <= 5.45
|   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |   |   |--- petal length (cm) > 5.45
|   |   |   |   |   |   |   |   |--- class: 2
|   |   |--- petal width (cm) > 1.75
|   |   |   |--- petal length (cm) <= 4.85
|   |   |   |   |--- sepal length (cm) <= 5.95
|   |   |   |   |   |--- class: 1
|   |   |   |   |   |--- sepal length (cm) > 5.95
|   |   |   |   |   |   |--- class: 2
|   |   |--- petal length (cm) > 4.85
|   |   |   |--- class: 2

```

```

TREE USING ENTROPY / INFORMATION GAIN:
|--- petal length (cm) <= 2.45
|   |--- class: 0
|--- petal length (cm) > 2.45
|   |--- petal width (cm) <= 1.75
|   |   |--- petal length (cm) <= 4.95
|   |   |   |--- petal width (cm) <= 1.65
|   |   |   |   |--- class: 1
|   |   |   |   |--- petal width (cm) > 1.65
|   |   |   |   |   |--- class: 2
|   |   |--- petal length (cm) > 4.95
|   |   |   |--- petal width (cm) <= 1.55
|   |   |   |   |--- class: 2
|   |   |   |   |--- petal width (cm) > 1.55
|   |   |   |   |   |--- petal length (cm) <= 5.45
|   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |   |--- petal length (cm) > 5.45
|   |   |   |   |   |   |   |--- class: 2
|   |--- petal width (cm) > 1.75
|   |   |--- petal length (cm) <= 4.85
|   |   |   |--- sepal length (cm) <= 5.95
|   |   |   |   |--- class: 1
|   |   |   |   |--- sepal length (cm) > 5.95
|   |   |   |   |   |--- class: 2
|   |--- petal length (cm) > 4.85
|   |   |--- class: 2

```

**COMPARISON:**

Both trees have the same structure.